

Uzair Abid

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RESEARCH INTERESTS

Natural Language Processing • Large Language Models and Agentic AI • LLM-based simulation of human behavior • Computer Vision for edge deployment • Efficient training and inference (PEFT, quantization) • Evaluation and reliability of LLM systems.

I am interested in building intelligent systems that understand language, learn from data, and act reliably in real-world settings. My goal for graduate study is to move from applied engineering toward rigorous research on how large language models can be evaluated, grounded, and used as instruments for studying human decision-making.

EDUCATION

Sir Syed University of Engineering and Technology

B.S. in Computer Engineering | CGPA: 3.58 / 4.00

Karachi, Pakistan

Oct 2020 – Jul 2024

- **Relevant coursework:** Artificial Intelligence, Operating Systems, Database Management Systems, Software Engineering, Object-Oriented Programming, Data Structures and Algorithms.
- **Final Year Project:** DevOps pipeline and scalable Kubernetes deployment platform, competitively selected for national funding by Ignite (National Technology Fund).
- University Merit Scholarship for five consecutive semesters (first position in class).

PUBLICATIONS & PREPRINTS

- [1] **U. Abid** and B. Lim. *Using Large Language Models to Study Consumer Behavior: Review Algorithm Effects on Restaurant Selection*. Manuscript in preparation, 2026. [\[In progress\]](#)
- [2] **U. Abid** et al. *Smart Shelf Advertising using Real-Time Product Segmentation and Interaction on Low-Cost Edge Devices*. Preprint, 2025. [\[Preprint PDF\]](#)

RESEARCH EXPERIENCE

LLM Agents for Consumer Behavior Research

Co-Author, with Dr. Boram Lim (formerly University of Kansas)

Remote

Ongoing

- Designing an agent-based simulation framework in which GPT-powered agents make restaurant-choice decisions, enabling controlled experiments on consumer behavior at a scale that is infeasible with human subjects.
- Implementing a factorial experimental design over review-presentation algorithms (highest-rated vs. most-recent) and restaurant positioning (high-end vs. casual) to isolate their effects on choice.
- Built bias-detection and dynamic review-supplementation components to reduce systematic drift in agent responses, plus structured behavioral logging for quantitative analysis of decision traces.

Smart Shelf Advertising on Low-Cost Edge Devices

Co-Author

Karachi, Pakistan

2024 – 2025

- Developed a real-time retail system that detects product pick-up events and triggers contextual video advertisements, running end to end on low-cost edge hardware.
- Trained a custom YOLOv8 segmentation model on a self-collected dataset of 3,500+ images, achieving **97.4% mAP@0.5** with real-time inference on constrained devices.
- Integrated AprilTag-based tracking and Arduino-controlled lighting into a single pipeline, validating robustness under varied shelf and lighting conditions.

Simulation Study on Edge-Cloud Traffic

Research Assistant to Engr. Muhammad Rehan Rasheed

Sir Syed University

2023

- Built a Java-based discrete simulation of edge-cloud environments to analyze traffic-management strategies and their effect on latency and throughput.

PROFESSIONAL EXPERIENCE

Wealthica — Canada (Remote)

Jul 2024 – Present

Head of Web Data Acquisition; previously Software Engineer

Promoted Jun 2025

- Architected an AI-powered auto-remediation platform (Claude API, GitHub Actions, RabbitMQ, SigNoz, Slack, ClickUp) that detects connector failures, scrubs PII, drafts pull requests, and applies path-restricted AI fixes through CI/CD, cutting manual triage time by **70%** and issue resolution time by **50%**.
- Built a browser-automation platform driven by a custom-trained ML model that parses web-page structure to automate data extraction, improving extraction efficiency by **35%** and reducing manual effort by **60%**.
- Engineered backend services in Node.js, TypeScript, MongoDB, Redis, Docker, and REST APIs powering scalable financial data pipelines.
- Eliminated **30%** of peak-hour request failures by profiling load patterns and introducing predictive pre-scheduling of jobs with AWS CDK.
- Led API integrations for fintech partners including Questrade and Flinks, covering secure authentication flows and optimized data access.

AutomateBoring — United States (Remote)

Jul 2023 – Jul 2024

Software Engineer, ML/AI & Backend

- Built retrieval-augmented generation (RAG) systems pairing embedding models with vector databases (Pinecone, FAISS) to ground LLM outputs in proprietary corpora, substantially reducing hallucination on knowledge-intensive tasks.
- Fine-tuned open-source LLMs with LoRA/QLoRA (PEFT) on curated instruction datasets, matching the quality of substantially larger models at a fraction of training and inference cost.
- Accelerated inference via 4-bit/8-bit quantization and vLLM-based serving, increasing throughput while reducing GPU memory footprint and latency in production.
- Developed computer-vision pipelines for object detection, segmentation (YOLOv8), and OCR in PyTorch, using mixed-precision and distributed (DDP) training to shorten experiment cycles.
- Established MLOps practice: experiment tracking (Weights & Biases), automated evaluation harnesses with LLM-as-judge, model versioning, and CI/CD, enabling reproducible experiments and dependable releases.

MyAskAI — United Kingdom (Contract, Remote)

Apr 2023 – Jul 2023

AWS DevOps Engineer

- Deployed a machine learning model on AWS as an end-to-end pipeline with model hosting, input pre-processing, and result handling using Lambda, S3, and API Gateway.
- Migrated a three-tier application from Render to AWS with CI/CD via CodePipeline, improving performance by **30%** and reducing hosting costs by **20%**.
- Instrumented observability with Datadog and CloudWatch, surfacing 15 latent issues and cutting user-reported incidents by **25%**.

Rehani Soko — United States (Contract, Remote)

Jan 2023 – May 2023

AWS DevOps Engineer

- Developed and optimized microservices for a real-estate platform using Lambda, S3, and DynamoDB, reducing latency by **18%**.
- Implemented CI/CD pipelines (CodePipeline, CodeBuild) with automated testing, deployment, and rollback across environments, cutting release times by **40%**.

TEACHING, MENTORING & ACADEMIC SERVICE

Stanford University — Code in Place

Section Leader

- Selected as a section leader for Stanford's CS106A-based global course; taught introductory Python to 10+ students across multiple countries. [\[Certificate\]](#)

North Dakota State University

Volunteer

- Assisted Dr. M. Zubair Malik in modernizing *CSCI 717 – Software Construction*, integrating AI-driven development techniques and cloud computing (AWS) to align the curriculum with current industry practice.

ICodeGuru

Trainer & Moderator

- Taught programming and algorithmic problem-solving (LeetCode) to students from underserved communities across Pakistan as part of a volunteer-run skills initiative.

Awards & Honors

- **Ignite Final Year Project Funding**, National Technology Fund, Pakistan, 2024
- **University Merit Scholarship**, five consecutive semesters for first position in class, 2020–2023
- **Top Rated Freelancer**, Upwork, 2022
- **Government of Sindh Academic Award** for A+ grade in Matriculation, 2017

Selected Projects

- **Resume AI Assistant**. Resume optimization system using GPT-4 with Pydantic-validated structured outputs; performs job-description-aware tailoring, gap analysis, cover-letter generation, and visual diffing of edits. [\[Code\]](#) • [\[Demo\]](#)
- **PDF Q&A Bot with Multi-LLM Backend**. Document QA over open-weight models (Phi-3.5, Qwen 2.5, Llama 3.2, Gemma 2) with 4-bit quantization and automatic CPU offload for low-memory deployment. [\[Code\]](#) • [\[Demo\]](#)
- **BlogPilot AI**. Multi-agent content pipeline in CrewAI with four specialized agents (research, outline, write, publish) and WordPress REST integration. Built for the Global AI Agents League [hackathon](#). [\[Code\]](#) • [\[Demo\]](#) • [\[Devpost\]](#)
- **Legal Hearing Summarizer**. Long-form legal transcript summarization system built for the [Azure AI Developer Hackathon](#). [\[Devpost\]](#)
- **AI Tutor**. Adaptive tutoring system (GPT-4o-mini) that modulates explanation complexity across six levels with real-time streaming and dynamic level adjustment. [\[Code\]](#) • [\[Demo\]](#)
- **Mental Health Discourse Classification**. A scikit-learn NLP pipeline classifying social-media posts related to stress, anxiety, and depression to characterize online mental-health discourse. [\[Code\]](#)
- **Flight Delay Prediction**. XGBoost model forecasting delays from historical flight and weather data, with extensive feature engineering and hyperparameter search. [\[Code\]](#)

Technical Skills

Languages	Python, TypeScript/JavaScript, Java, SQL, Bash
ML & AI	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, PEFT (LoRA/QLoRA), vLLM, LangChain, CrewAI, OpenCV, YOLOv8, Weights & Biases
LLM Systems	RAG, vector databases (Pinecone, FAISS), quantization (4/8-bit), evaluation harnesses & LLM-as-judge, agentic workflows
Backend	Node.js, Express.js, Nest.js, FastAPI, Django, Flask, MongoDB, Redis, DynamoDB, REST APIs
Cloud & DevOps	AWS (Lambda, S3, EC2, DynamoDB, API Gateway, CDK, CodePipeline), Docker, Kubernetes, GitHub Actions, Datadog, SigNoz

Certifications

- AWS Certified Cloud Practitioner, Amazon Web Services, Apr 2023 [\[Verify\]](#)
- Hands-on Introduction to Linux Commands and Shell Scripting, IBM, Mar 2023 [\[Verify\]](#)
- Node.js: The Complete Guide (MVC, REST APIs, GraphQL, Deno), Nov 2022 [\[Verify\]](#)
- Programming for Everybody (Python), University of Michigan, Jul 2020 [\[Verify\]](#)

References available on request.